

**Solutions of Part 2 Exam: Mathematics for
Microeconomics**

1. (20 marks) Consider the following quadratic form:

$$Q(\mathbf{x}) = 2x_1^2 + 4x_3^2 + 4x_1x_2 - 2x_1x_3$$

- (a) (4 marks) Identify the *symmetric matrix of coefficients* A associated with this quadratic form. That is

$$Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$$

Find all the leading principal submatrices of A .

Solutions: The matrix of coefficients is:

$$A = \begin{pmatrix} 2 & 2 & -1 \\ 2 & 0 & 0 \\ -1 & 0 & 4 \end{pmatrix}$$

Its leading principal submatrices are:

$$A_3 = \begin{pmatrix} 2 & 2 & -1 \\ 2 & 0 & 0 \\ -1 & 0 & 4 \end{pmatrix} \quad A_2 = \begin{pmatrix} 2 & 2 \\ 2 & 0 \end{pmatrix} \quad A_1 = (2)$$

- (b) (8 marks) Define what it means for the quadratic form $Q(\mathbf{x})$ to be *positive definite*. How would you be able to assess, in terms of the properties of A , whether $Q(\mathbf{x})$ is positive definite?

Solutions: A quadratic form is positive definite if $Q(\mathbf{x}) > 0$ for any $\mathbf{x} \neq \mathbf{0}$. The quadratic form is positive definite if all the leading principal minors of A are positive.

- (c) (8 marks) Is the quadratic form $Q(\mathbf{x})$ *positive definite*, *negative definite*, *positive semi-definite*, *negative semi-definite* or *indefinite*?

Solutions: $Q(\mathbf{x})$ is indefinite since A is indefinite since the third and the first leading minors have opposite signs:

$$|A_1| = 2 > 0, \quad |A_2| = \begin{vmatrix} 2 & 2 \\ 2 & 0 \end{vmatrix} = -4 < 0,$$

$$|A_3| = \begin{vmatrix} 2 & 2 & -1 \\ 2 & 0 & 0 \\ -1 & 0 & 4 \end{vmatrix} = -16 < 0.$$

2. (40 marks) For some constant $k \in (0, 2)$ consider the function defined on $\mathbb{R}^2$:

$$f(x, y) = k(xy + x + y) - x^2 - y^2$$

- (a) (8 marks) Find the *critical points* of $f(x, y)$.

Solutions: The critical point of function $f(x, y)$ is

$$x^* = y^* = \frac{k}{2 - k}$$

and it is the solution to FOC

$$\frac{df(x, y)}{dx} = k(y + 1) - 2x = 0 \quad (1)$$

$$\frac{df(x, y)}{dy} = k(x + 1) - 2y = 0 \quad (2)$$

- (b) (8 marks) Are these critical points *local maximizers*, *local minimizers* or *neither*?

Solutions: The critical point $x(k) = y(k) = k/(2 - k)$ of the function $f(x, y)$ is a local maximizer since the sufficient second order condition for a local minimizer is satisfied at such point:

$$H = \frac{d^2 f(x(k), y(k))}{d(x, y)^2} = \begin{bmatrix} -2 & k \\ k & -2 \end{bmatrix} \quad (3)$$

Note that the Hessian above is negative definite as leading principal minors alternate in signs with the appropriate pattern, $H_1 < 0$ and $H_2 > 0$.

- (c) (8 marks) Are these critical points *global maximizers*, *global minimizers* or *neither*?

Solutions: The critical point $x(k) = y(k) = k/(2 - k)$ of the function $f(x, y)$ is also a global maximizer since

$$\frac{d^2 f(x, y)}{d(x, y)^2} = \begin{bmatrix} -2 & k \\ k & -2 \end{bmatrix}$$

is negative definite for any (x, y) pair as $k < 2$.

- (d) (8 marks) Is $f(x, y)$ *concave*, *convex*, *quasi-concave*, *quasi-convex* or *neither of the above*?

Solutions: The concavity of function follows from the solution to part c. Thus, the function is also quasi-concave. The function is clearly not convex as it is strictly concave on the entire domain. Finally the function is not quasi-convex since the points $(1, -1)$ and $(-1, 1)$ sit on the same level curve $f(x, y) = -(k + 2)$ but their convex combination $1/2(1, -1) + 1/2(-1, 1) = (0, 0)$ sits on a higher level curve $f(x, y) = 0$.

- (e) (8 marks) Denote the *global maximizer* of $f(x, y)$ by $(x(k), y(k))$. Use the envelope theorem to show how *maximum value function* responds to a marginal change in k . Also show how the *global maximizer* $(x(k), y(k))$ is affected by marginal changes in k .

Solutions: By the envelope theorem we know that:

$$\begin{aligned} \frac{d}{dk} f(x(k), y(k), k) &= \frac{\partial}{\partial k} f(x(k), y(k), k) = \\ &= x(k)y(k) + x(k) + y(k) \\ &= \frac{4k - k^2}{(2 - k)^2} > 0 \end{aligned}$$

Furthermore since $x(k) = y(k) = k/(2 - k)$:

$$x'(k) = y'(k) = \frac{2}{(2 - k)^2} > 0$$

3. (40 marks) For $I > 0$, consider the problem:

$$\begin{aligned} \max_{x,y} \quad & F(x, y) = -e^{-xy} \\ \text{s.t.} \quad & x + 2y \leq I \\ & x \geq 0 \\ & y \geq 0 \end{aligned} \tag{4}$$

where e^z denotes the exponential function evaluated at z .

- (a) (8 marks) Establish whether the objective function and the constraints are respectively concave, convex, or neither. Which conditions on the concavity/convexity of such functions would suffice to guarantee that the Lagrangian is concave?

Solutions: The objective function is indefinite for any $x, y \geq 0$:

$$d^2 F(x, y) = \begin{pmatrix} -y^2 e^{-xy} & (1 - xy)e^{-xy} \\ (1 - xy)e^{-xy} & -x^2 e^{-xy} \end{pmatrix}$$

since $|d^2 F(x, y)| = (2xy - 1)e^{-xy}$, $-x^2 e^{-xy} \leq 0$ and $-y^2 e^{-xy} \leq 0$

Therefore it is neither concave nor convex. The constraints are obviously linear (both convex and concave). If the objective function were concave and the constraints were convex, the Lagrangian would have been concave, since it is a sum of concave functions.

- (b) (8 marks) Set up the constrained optimization problem using the Lagrangian function. Can we solve this optimization problem using the Kuhn-Tucker method? Are the Kuhn-Tucker constraint qualifications satisfied? Is it possible to solve the same optimization problem with the Lagrangean method? Are the Lagrange constraint qualifications satisfied?

Solutions: The Lagrangian of the problem is

$$\mathcal{L}(x, y, \lambda_1, \lambda_2, \lambda_3) = -e^{-xy} - \lambda_1(x + 2y - I) + \lambda_2 x + \lambda_3 y$$

Notice first that a simple contradiction argument proves that if $I > 0$ the optimal solution cannot have $x = 0$ and/or $y = 0$, as we could increase the value of the program by setting $x = y = I/3$.

The Lagrangian of the problem then becomes:

$$\mathcal{L}(x, y, \lambda_1, \lambda_2, \lambda_3) = -e^{-xy} - \lambda_1(x + 2y - I)$$

The constraint set has a non-empty interior thus we can adopt the Kuhn-Tucker approach to solve the problem. Moreover, the gradient of the constraint $g(x, y) = x + 2y - I$ is:

$$\nabla g(x, y) = (1, 2)$$

therefore constraint qualifications for the Lagrange method are satisfied for every pair (x, y) such that $x + 2y \leq I$. In other words, we can use the Lagrangian method to solve the constrained maximization problem.

- (c) (8 marks) Write down the necessary first order conditions for the solution of the constrained optimization problem.

Solutions: The necessary first order conditions for the solution of the constrained maximization problem are then:

$$\begin{aligned} 0 &= ye^{-xy} - \lambda_1 \\ 0 &= xe^{-xy} - 2\lambda_1 \\ 0 &= \lambda_1(x + 2y - I) \\ 0 &\geq x + 2y - I \\ 0 &\leq \lambda_1 \end{aligned}$$

- (d) (4 marks) Are these conditions also *sufficient* to characterize an optimum in this problem?

Solutions: The necessary first order conditions for the solution of the constrained maximization problem are not sufficient for a maximum since the objective function is not concave.

- (e) (4 marks) Solve the constrained optimization problem in terms of I .

Solutions: The solution to the constraint optimization problem must be one of the two solution to the necessary conditions above. In particular, it is the one in which $\lambda_1 > 0$ and it satisfies $\lambda_1 = (I/4)e^{-I^2/8}$ and $(x^*(I), y^*(I)) = (I/2, I/4)$.

- (f) (8 marks) Let $x^*(I)$ and $y^*(I)$ be the solution of this problem. Find the slope of such functions with respect to I . Moreover, find how the maximum value function is affected by a change in I .

Solutions: Since the bordered Hessian is negative definite a the global optimum, the implicit function theorem applies. Thus, it is immediate to observe that:

$$\frac{\partial}{\partial I}x^*(I) = \frac{1}{2} \text{ and } \frac{\partial}{\partial I}y^*(I) = \frac{1}{3}.$$

Finally, observe that by the Envelope theorem:

$$\begin{aligned} \frac{d}{dI}F(x^*(I), y^*(I)) &= \\ &= \frac{\partial}{\partial x}F(x^*(I), y^*(I))\frac{\partial}{\partial I}x^*(I) + \frac{\partial}{\partial y}F(x^*(I), y^*(I))\frac{\partial}{\partial I}y^*(I) = \\ &= \lambda_1 \left[\frac{\partial}{\partial x}g(x^*(I), y^*(I), I)\frac{\partial}{\partial I}x^*(I) + \frac{\partial}{\partial y}g(x^*(I), y^*(I), I)\frac{\partial}{\partial I}y^*(I) \right] = \\ &= \lambda_1 = (I/4)e^{-I^2/8} \end{aligned}$$

for $g(x, y, I) = x + 2y - I$.